A determinant dichotomy for quadratic zero-one systems: steady-state parity, boundedness, and finite-time escape

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Abstract

We study three-variable quadratic systems whose coefficient of a monomial in the ith equation is negative when that monomial contains x_i, and positive otherwise. These are precisely the steady-state sign constraints of quadratic zero-one reaction networks. For systems with all 21 admissible coefficients nonzero, a 3×3 matrix formed only from the quadratic coefficients gives a global dichotomy. A negative determinant implies an absorbing box and Brouwer degree -1; a positive determinant implies finite-time escape from an open set and degree zero. Consequently, when all positive steady states are nondegenerate, their number is respectively in $\{1, 3, 5\}$ or $\{0, 2, 4\}$. The bound four in the positive-determinant region extends to sparse systems and is sharp. We give an elementary seven-reaction witness with integer rates and show explicitly that its two stable steady states coexist with finite-time escape. An eight-reaction network attains five positive steady states, the Bernstein bound, three of them stable; its quadratic determinant is negative, as the dichotomy requires. Both extreme examples have elementary proofs, and both are formally verified in Lean 4.

1 Introduction and scope

The number of positive steady states of a mass-action system measures a different aspect of its dynamics from the stability of any one state. For three-species quadratic zero-one networks, Tang and Zhang prove determinant alternation and, by computer classification, an upper bound of three nondegenerate positive steady states when there are six reactions [2]. The support bound from Bernstein's theorem is five, independently of the number of reactions. The gap between these bounds raises two distinct questions: what changes when more reactions are allowed, and what controls the behaviour at infinity?

A small example shows what is at stake. The seven reactions $X_1 + X_2 \to 0$ (rate 7), $X_1 + X_3 \to X_2 + X_3$, $X_3 \to X_1$, $X_1 + X_2 \to X_3$, $X_2 + X_3 \to X_1 + X_2 + X_3$, $0 \to X_2$ (rate 1 each) and $X_1 \to X_1 + X_3$ (rate 3) give, with concentrations (x, y, z), the mass-action system

$$f = (-8xy - xz + z + yz,\ -8xy + xz + 1,\ xy - z + 3x).$$

It has exactly four positive steady states, two of them stable (Section 5). It also has solutions that leave every bounded set in finite time: from any start with $18x$, $9y$ and z all above 324, the solution ceases to exist before time one. Both facts are governed by the quadratic terms. Record, for each equation in turn, its coefficients of yz, xz and xy:

$$B = \begin{pmatrix} 1 & -1 & -8 \\ 0 & 1 & -8 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det B = 1 > 0.$$

For systems in which every admissible coefficient is present, the main theorem says that a positive determinant brings exactly this combination: an even number of steady states, at most four, together with escape to infinity. A negative determinant brings the opposite: every trajectory enters one bounded box, and the number of steady states is odd.

Species and reaction minimality for zero-one networks was also studied in [3]. We connect the counting and asymptotic questions through the quadratic coefficient matrix. The result is independent of reaction-count enumeration. Its two substantive parts are a constructive dynamical criterion and an exact degree computation. Only the stability-count corollary uses the determinant criterion of [2]. The sharp four-state example also supplies an upper witness for the least reaction count, which is seven using the six-reaction theorem of [2].

The bound five is attained. Section 6 gives an eight-reaction network with exactly five positive steady states, three stable and two unstable, and proves this with one rational curve and one quintic. Read together, the two examples locate the count of three in [2]: it is a consequence of having six reactions. One more reaction gives four, and two more give the largest number any network in the class can have.

The assumptions matter. The global dichotomy and parity assertion are proved for the strict coefficient class and away from a determinant-zero hypersurface. They do not prove a universal upper bound of four for all networks. Networks with few reactions are sparse: the seven-reaction example fills 10 of the 21 coefficient slots. For sparse systems the bound four and the escape criterion survive (Corollary 2.2), while parity can fail (Section 9 gives a sparse four-state network with $\Delta < 0$). Exact computations verify the examples; they are not used as evidence for the general theorem.

2 The coefficient matrix and the main result

A reaction $\sum_i \mu_i X_i \to \sum_i \nu_i X_i$ with rate k contributes $k(\nu_i - \mu_i)\, x^\mu$ to f_i. In a zero-one network $\nu_i - \mu_i$ is -1, 0 or 1, and it is -1 only when X_i is a reactant, that is, only when the monomial x^μ contains x_i. In the example above, $X_1 + X_2 \to X_3$ contributes $-xy$ to f_1 and f_2 and $+xy$ to f_3. Reactants of molecularity at most two give at most the seven monomials $1, x_i, x_i x_j$, and the sign rule fixes the sign of each coefficient. For $\{i,j,k\} = \{1,2,3\}$ this gives the general form

$$f_i(x) = a_i - b_i x_i + c_{ij} x_j + c_{ik} x_k - d_{ij} x_i x_j - d_{ik} x_i x_k + e_i x_j x_k. \tag{1}$$

Throughout the main theorem all $a_i, b_i, c_{ij}, d_{ij}, e_i$ are strictly positive. This will be called the *strict class*. Let Q be the homogeneous quadratic part of f, and define

$$B = \begin{pmatrix} e_1 & -d_{13} & -d_{12} \\ -d_{23} & e_2 & -d_{21} \\ -d_{32} & -d_{31} & e_3 \end{pmatrix}, \qquad \Delta = \det B. \tag{2}$$

The ordering of its columns is x_2x_3, x_1x_3, x_1x_2; for the example of Section 1 this is the matrix displayed there. Thus

$$Q(x) = x_1 x_2 x_3 B \begin{pmatrix} x_1^{-1} \\ x_2^{-1} \\ x_3^{-1} \end{pmatrix}. \tag{3}$$

A steady state is nondegenerate if $\det Df(x) \neq 0$. The system is *regular in the positive orthant* if all its positive steady states are nondegenerate. In the statements below, an absorbing box means one fixed bounded box that every positive trajectory eventually enters and subsequently stays in.

Theorem 2.1 (Determinant dichotomy)**.** *Let f belong to the strict class and suppose $\Delta \neq 0$.*

1. *If $\Delta < 0$, every positive solution is global and there is an absorbing box. The degree of f on a bounded open set containing all its positive zeros is -1. If the system is regular, its number N of positive steady states belongs to $\{1, 3, 5\}$ and*

$$\sum_{f(p)=0,\ p>0} \operatorname{sgn} \det Df(p) = -1.$$

2. *If $\Delta > 0$, there is a positive vector w and $R > 0$ such that every initial point with $x_i > Rw_i$ for all i has finite maximal forward existence time. The degree of f on a bounded open set containing all its positive zeros is zero. If the system is regular, $N \in \{0, 2, 4\}$ and the signed sum above is zero.*

All degree domains here have closure inside the positive orthant and contain all positive zeros, so the asserted degree is unambiguous by excision.

Corollary 2.2 (Sparse extension and sharpness)**.** *Allow any coefficients in (1) to be zero, while retaining their prescribed weak signs. If $\det B > 0$, there are at most four nondegenerate positive steady states, whether or not other positive steady states are degenerate. There is also an open set of positive initial points with finite maximal forward existence time. The bound four is attained by a seven-reaction quadratic zero-one network and by systems in the strict class.*

Corollary 2.3 (Four states versus five)**.** *Let $\Delta \neq 0$. A regular strict system with exactly four positive steady states has an open set of finite-time escaping trajectories. A regular strict system with five positive steady states, if one exists, must lie in the absorbing-box region $\Delta < 0$.*

Corollary 2.4 (Reduction of the five-state existence problem)**.** *There exists a weak sign-pattern system with five nondegenerate positive steady states if and only if there exists a regular strict system with five positive steady states and an absorbing box.*

Proof. Perturb a putative weak five-state example into the strict class. Choose the perturbation generically, with all positive zeros nondegenerate and $\Delta \neq 0$. The five original states persist, and Bernstein's bound excludes additional nondegenerate positive states. Theorem 2.1 excludes $\Delta > 0$, so $\Delta < 0$ and there is an absorbing box. The converse is immediate. □

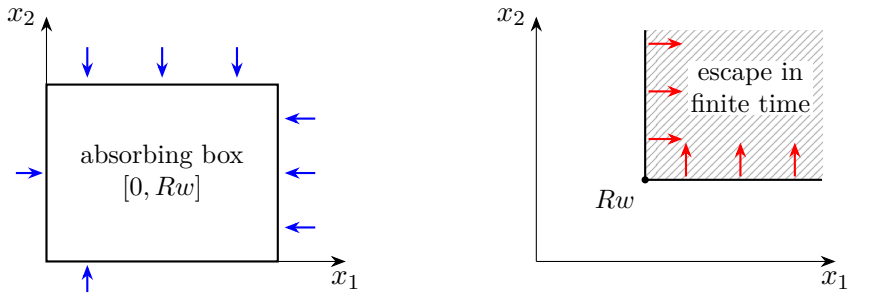
Corollary 2.5 (Stability count, using Tang–Zhang)**.** *Under the regularity assumption of Theorem 2.1, the numbers of stable and unstable positive steady states are equal when $\Delta > 0$. When $\Delta < 0$, there is exactly one more stable than unstable positive steady state.*

Proof. Each strict sign-pattern system is realised by a full-dimensional quadratic zero-one network: a positive monomial in f_i is realised by adding X_i to its source complex, and a negative monomial by deleting X_i. The nonzero inflows and outflows make the realisation full-dimensional. By [2, Lemma 3], a nondegenerate positive steady state is stable exactly when its determinant is negative; its proof uses [4, Lemma 6.6(ii)] (Lemma 6.5(ii) in the arXiv version), which holds for every zero-one network and so for these realisations. Substituting this criterion into the signed degree sum proves the claim. In particular, five states would mean three stable and two unstable states. □

In words, (3) says that the quadratic part at x is $x_1x_2x_3$ times B applied to the vector of reciprocals. Along a ray $x = rw$ with $w > 0$ fixed, $Q(rw) = r^2\, w_1w_2w_3\, B(1/w)$, so the signs of Bv at $v = 1/w$ decide whether the quadratic terms push outward or inward along that ray.

The proof has three steps, pictured in Figure 1. Section 3 turns the sign of Bv into explicit barriers: when $Bv < 0$ for some $v > 0$, the upper faces of every large box of shape $1/v$ point inward;

when $Bv > 0$, the lower faces of a far upper orthant point outward and solutions beyond them grow at least quadratically, hence escape in finite time. Section 4 computes the Brouwer degree of f over a domain holding all positive zeros. On the absorbing box a straight homotopy to $c - x$ gives -1. When $\Delta > 0$ a homotopy inside the region $\Delta > 0$ ends at a symmetric system with no positive zeros, so the degree is 0; a compactness lemma keeps zeros away from the boundary along the way. Parity follows, and Bernstein's bound of five isolated zeros turns parity into the two lists.



$\Delta < 0$: $Bv < 0$ for some $v > 0$ $\qquad$ $\Delta > 0$: $Bv > 0$ for $v = B^{-1}\mathbf{1} > 0$

Figure 1: The dichotomy in a two-dimensional slice, with $w = 1/v$. Left: the upper faces of the box point inward and the lower faces, where $f_i \geq a_i > 0$, point inward too, so every positive solution enters the box. Right: on the lower faces of the upper orthant beyond Rw the field points outward, and solutions starting there leave every bounded set in finite time.

3 Matrix alternatives and rectangular barriers

We give the estimates explicitly because off-diagonal derivatives of f need not have a fixed sign. A global comparison theorem for cooperative vector fields is therefore unnecessary.

Lemma 3.1 (Matrix alternative)**.** *For a matrix B with positive diagonal and strictly negative off-diagonal entries: if $\det B > 0$, then $B^{-1}\mathbf{1} > 0$; if $\det B < 0$, there exists $v > 0$ with $Bv < 0$. For weakly negative off-diagonal entries and nonnegative diagonal, $\det B > 0$ still implies the first conclusion and a positive diagonal.*

Proof. Write $B = D(I - A)$ with D its positive diagonal and A entrywise nonnegative with zero diagonal. In the strict case A is irreducible. Its characteristic polynomial is $t^3 - St - T$, where $S, T > 0$, so it has exactly one positive real root, its Perron root ρ. The remaining two eigenvalues are either both negative or a conjugate pair; hence $\det(I - A)$ has the sign of $1 - \rho$. If $\det B > 0$, then $\rho < 1$ and $(I - A)^{-1} = \sum_{n \geq 0} A^n \geq I$. If $\det B < 0$, a positive Perron vector satisfies $Bv = (1 - \rho)Dv < 0$.

For the weak case expand the determinant, writing the off-diagonal magnitudes as $m_{ij} \geq 0$ and the diagonal as $q_i \geq 0$:

$$\begin{aligned}\det B = {} & q_1q_2q_3 - q_1m_{23}m_{32} - q_2m_{13}m_{31} - q_3m_{12}m_{21} \\ & - m_{12}m_{23}m_{31} - m_{13}m_{21}m_{32}.\end{aligned}$$

A positive determinant forces each $q_i > 0$ and all principal two-by-two minors positive. The diagonal cofactors are therefore positive and all off-diagonal cofactors in the adjugate nonnegative. Thus $B^{-1}\mathbf{1} > 0$. □

Proposition 3.2 (An inward quadratic direction gives an absorbing box)**.** *For a strict system, if $Bv < 0$ for some $v > 0$, there is an absorbing box.*

Proof. Set $w_i = 1/v_i$, so $Q_i(w) < 0$. For a fixed i put

$$U_i = d_{ij}w_iw_j, \quad V_i = d_{ik}w_iw_k, \quad W_i = e_iw_jw_k, \quad \delta_i = \min\{U_i, V_i, (U_i + V_i - W_i)/2\} > 0.$$

The bilinear function $-U_iu - V_iv + W_iuv + \delta_i(u + v)$ is nonpositive at all four corners of $[0, 1]^2$, and consequently throughout that square. On the upper face $x_i = rw_i$, $x_j = rw_ju$, $x_k = rw_kv$, with $0 \le u, v \le 1$, it follows that

$$f_i(x) \le a_i - b_irw_i - \delta_ir^2(u + v) + r(c_{ij}w_ju + c_{ik}w_kv).$$

Choose

$$R_1 = \max_i \frac{\max(c_{ij}w_j, c_{ik}w_k)}{\delta_i}, \quad A_0 = \max_i a_i/w_i, \quad b_0 = \min_i b_i > 0.$$

For $r(x) = \max_i x_i/w_i \ge R_1$, the upper Dini derivative obeys

$$D^+r \le A_0 - b_0r.$$

Thus every positive solution is bounded, cannot cease to exist in finite time, and eventually enters any box $[0, Rw]$ with $R > \max\{R_1, 2A_0/b_0\}$. Such a box is positively invariant. The lower faces point strictly inward because $a_i > 0$ and all remaining terms in $f_i|_{x_i=0}$ are nonnegative. This also ensures forward invariance of the positive orthant. □

Proposition 3.3 (An outward quadratic direction gives finite-time escape)**.** *In the weak sign class, if $Bv > 0$ for some $v > 0$, an open upper orthant of initial points has finite maximal forward existence time.*

Proof. Set $w_i = 1/v_i$ and $\gamma = \min_i Q_i(w)/w_i > 0$. On a lower face $x_i = rw_i$, $x_j = rw_ju$, $x_k = rw_kv$ of the upper orthant, $u, v \ge 1$. With U_i, V_i, W_i as above, $W_i > U_i + V_i$. Therefore $-U_iu - V_iv + W_iuv$ is increasing in each variable on $[1, \infty)^2$, and is bounded below by $Q_i(w) > 0$. The affine terms give

$$\frac{f_i(x)}{w_i} \ge \frac{Q_i(w)}{w_i}r^2 - b_ir.$$

Consequently, for $r(x) = \min_i x_i/w_i$ and $b_* = \max_i b_i$, its lower Dini derivative satisfies $D_+r \ge \gamma r^2 - b_*r$. If $r(0) > 2b_*/\gamma$, then r increases and $D_+r \ge \gamma r^2/2$. Integrating the reciprocal inequality shows that the maximal forward existence time is at most $2/(\gamma r(0))$. A polynomial ODE with a bounded solution extends past any finite time; hence finite maximal existence time is finite-time escape in norm. No claim about an exact escape profile or divergence of each individual coordinate is needed. □

4 Degree and the sharp bound

The preceding barriers prove the dynamical assertions in the main theorem. We now compute the degree. Only the positive-determinant case requires a homotopy argument.

Lemma 4.1 (Uniform compactness of positive zeros)**.** *For a compact family of strict systems with $\det B \ne 0$ throughout, all positive zeros lie in a common compact subset of the positive orthant.*

Proof. Let every coefficient magnitude lie in $[\eta, M]$ with $\eta > 0$. At a positive zero,

$$x_i = \frac{a_i + c_{ij}x_j + c_{ik}x_k + e_i x_j x_k}{b_i + d_{ij}x_j + d_{ik}x_k} \le K \min\{1 + x_j, 1 + x_k\}, \qquad K = M/\eta.$$

If one coordinate tends to infinity along a sequence of positive zeros, all three tend to infinity and their ratios stay bounded above and below. Let $H = \max_i x_i$. Then H/x_i lies in a fixed compact subset of $(0, \infty)$. Dividing $f = 0$ by $x_1x_2x_3$ and multiplying by H gives

$$B(H/x_1, H/x_2, H/x_3)^T = o(1).$$

A subsequence yields a nonzero positive vector in the kernel of a limiting B, contradicting nonsingularity. Hence positive zeros are uniformly bounded above. They cannot approach a coordinate hyperplane, since on a bounded set the limit of f_i at $x_i = 0$ is at least $\eta > 0$. □

Lemma 4.2 (Degree in the positive-determinant region)**.** *Every strict system with* $\det B > 0$ *has degree zero on a bounded open domain containing all its positive zeros.*

Proof. Positive multiplication of equations preserves their zeros and degree signs. Normalise the diagonal of B to one, obtaining $B = I - A$ with A positive off the diagonal and $\rho(A) < 1$. Scale A down through positive multiples until its maximum row sum is less than $1/4$. Then interpolate it to the matrix with all off-diagonal entries $1/10$; each intermediate row sum is less than one, so $\rho < 1$ throughout. Keep all lower-degree coefficients positive, and interpolate them to reach the symmetric system

$$g_i = 1 - x_i + \tfrac{1}{10}(x_j + x_k) - \tfrac{1}{10}x_i(x_j + x_k) + x_j x_k.$$

These paths stay inside the strict positive-determinant region. By Lemma 4.1, a common bounded degree domain exists along the entire path, so the degree is constant. At a positive zero of g,

$$g_i - g_j = (x_j - x_i)\tfrac{11}{10}(1 + x_k) = 0,$$

which forces all coordinates equal. But the diagonal equation is $1 - \frac{4}{5}t + \frac{4}{5}t^2 = 0$, whose discriminant is negative. The endpoint has no positive zeros and therefore degree zero. □

Proof of Theorem 2.1. If $\Delta < 0$, Lemma 3.1 and Proposition 3.2 give a large invariant box with strictly inward upper faces. On its lower faces $f_i > 0$. The straight homotopy from f to $h(x) = c - x$, with c in the box's interior, has no boundary zeros: on each face its normal component keeps the same strict sign. Hence the degree is $\deg(h) = -1$. By excision it can also be computed on a domain with closure in the positive orthant. If $\Delta > 0$, use Lemma 4.2 for the degree and Proposition 3.3, with $v = B^{-1}\mathbf{1}$, for finite-time escape.

For regular systems the zero set is compact and discrete, hence finite, and the degree equals the sum of Jacobian determinant signs. Reduction modulo two therefore gives odd parity for $\Delta < 0$ and even parity for $\Delta > 0$. The Newton polytope is

$$P = [0, 1]^3 \cap \{u_1 + u_2 + u_3 \le 2\}.$$

It is a unit cube minus a tetrahedron of volume $1/6$, so $3!\,\mathrm{Vol}\,P = 5$. Bernstein's bound [6], or its perturbative upper-bound form for nondegenerate roots, gives at most five isolated nondegenerate positive zeros. Combining this bound with parity proves the stated lists. □

Proof of the sparse extension in Corollary 2.2. The dynamical claim follows from Lemma 3.1 and Proposition 3.3. For the root bound, suppose five nondegenerate positive zeros existed. They persist under all sufficiently small coefficient perturbations. Perturb into the strict class while preserving $\det B > 0$, and choose a generic perturbation so that all positive zeros are nondegenerate. Such perturbations exist densely, since the singular systems lie in the algebraic discriminant locus; independent constant perturbations also give transversality directly. The strict system would then have at least five positive zeros, contradicting the even bound four. Sharpness is proved in the next section. □

5 A seven-reaction witness and an explicit escape region

Consider the reactions

Reaction	Rate	Reaction	Rate
$X_1 + X_2 \to 0$	7	$X_2 + X_3 \to X_1 + X_2 + X_3$	1
$X_1 + X_3 \to X_2 + X_3$	1	$0 \to X_2$	1
$X_3 \to X_1$	1	$X_1 \to X_1 + X_3$	3
$X_1 + X_2 \to X_3$	1		

Quadratic refers to reactant degree, as in [2]; a product may contain three species. The stoichiometric rank is three, as witnessed by the vectors $(0,1,0), (1,0,-1), (0,0,1)$.

Writing concentrations as (x, y, z) gives

$$f = (-8xy - xz + z + yz,\ -8xy + xz + 1,\ xy - z + 3x). \tag{4}$$

The identity $f_2 + 8f_3 = x(z + 24) - (8z - 1)$ shows that any positive zero has

$$z > 1/8, \qquad x = \frac{8z-1}{z+24}, \qquad y = \frac{z^2+3}{8z-1}.$$

The last two equations then vanish identically, and the first is

$$f_1 = \frac{P(z)}{(z+24)(8z-1)}, \qquad P(z) = z^4 - 96z^3 + 218z^2 - 145z + 24.$$

Exactly,

$$\big(P(1/8), P(1/2), P(1), P(2), P(96)\big) = \big(37249/4096, -95/16, 2, -146, 1995192\big).$$

Four disjoint sign changes (Figure 2) prove that all four roots are simple and larger than $1/8$. They give exactly four positive steady states. The minor $\det \partial(f_2, f_3)/\partial(x, y) = x(z + 24) > 0$ yields, by differentiation along the eliminated curve,

$$\det Df = \frac{P'(z)}{z+24} \quad \text{at every steady state.}$$

This proves nondegeneracy and signs $-, +, -, +$ independently of determinant alternation. Direct rational Routh–Hurwitz certificates verify two stable states. For the characteristic coefficients b_1, b_2, b_3 and $H_2 = b_1 b_2 - b_3$, the first state has enclosures $(26, 27), (6, 7), (2, 3), (179, 180)$ and the third has $(9, 10), (13, 14), (1, 2), (125, 126)$, respectively.

The behaviour at infinity is equally explicit:

$$B = \begin{pmatrix} 1 & -1 & -8 \\ 0 & 1 & -8 \\ 0 & 0 & 1 \end{pmatrix}, \quad B^{-1}\mathbf{1} = (18, 9, 1)^T, \quad w = (1/18, 1/9, 1)^T.$$

Here $Q_i(w) = 1/162$, $\gamma = 1/162$, and $b_* = 1$. The proof of Proposition 3.3 gives

$$r_0 = \min\{18x(0), 9y(0), z(0)\} > 324 \quad \Longrightarrow \quad T_{\max} \leq 324/r_0 < 1. \tag{5}$$

Thus two attracting equilibria coexist with an open region of finite-time escape. Local multistability alone does not imply global boundedness.

Using [2, Theorem 2], the least number of reactions for four nondegenerate positive steady states is seven. The theorem excludes six. It is proved by real root classification, which certifies counts for rate vectors off a discriminant hypersurface, but it bounds every rate vector: four nondegenerate states at one rate vector would persist on an open set of rate vectors, and an open set meets the complement of a hypersurface. If a smaller network had four such states, add unused quadratic zero-one reactions at sufficiently small positive rates until there are six; nondegenerate positive states persist, giving a contradiction. This lower-bound argument relies on the cited computer-assisted theorem; the witness and escape proof do not.

6 Five steady states with eight reactions

Consider the reactions

Reaction	Rate	Reaction	Rate
$X_1 \to X_1 + X_3$	5	$X_2 + X_3 \to X_1$	2/5
$X_1 \to X_2$	1	$X_3 \to 0$	1/2
$X_1 + X_3 \to X_1 + X_2 + X_3$	1	$0 \to X_1 + X_2$	1/2500
$X_1 + X_2 \to X_2$	1	$X_2 \to X_1$	1/100000

The network is quadratic zero-one with stoichiometric rank three. With concentrations (x, y, z) the mass-action system is

$$\begin{aligned} f_1 &= \tfrac{1}{2500} - x - xy + \tfrac{2}{5}yz + \tfrac{1}{100000}y, \\ f_2 &= \tfrac{1}{2500} + x + xz - \tfrac{2}{5}yz - \tfrac{1}{100000}y, \\ f_3 &= 5x - \tfrac{1}{2}z - \tfrac{2}{5}yz. \end{aligned} \tag{6}$$

Theorem 6.1. *The system (6) has exactly five positive steady states. All five are nondegenerate, with* $\det Df$ *of signs* $-, +, -, +, -$ *in increasing* z*; three are asymptotically stable and two are unstable.*

Proof. The equation $f_3 = 0$ is linear in x and gives $50x = z(5 + 4y)$. Substituting this into f_2 leaves an expression linear in y:

$$100000\, f_2 = y\, D(z) + N(z), \qquad D(z) = 8000z^2 - 32000z - 1, \quad N(z) = 10000z^2 + 10000z + 40.$$

At a positive steady state $N(z) > 0$, so $D(z) \neq 0$ and

$$y = Y(z) = -\frac{N(z)}{D(z)}, \qquad x = X(z) = \frac{z\,(5 + 4Y(z))}{50}.$$

Conversely, these two formulas make f_2 and f_3 vanish identically wherever $D(z) \neq 0$, so the curve $f_2 = f_3 = 0$ is rational with parameter z. Substituting into the remaining equation gives

$$f_1(X(z), Y(z), z) = -\frac{E(z)}{1250\, D(z)^2},$$

$$E(z) = 40000000000z^5 - 110031000000z^4 + 50416250000z^3 - 782738125z^2 + 101000z - 1.$$

Positive steady states therefore correspond to the roots $z > 0$ of E at which X and Y are positive. The exact values

z	0	$\frac{1}{20000}$	$\frac{1}{1000}$	$\frac{1}{10}$	1	3
$E(z)$	-1	$\frac{335912964971}{160000000000}$	$-\frac{316215933}{500000}$	$\frac{127983471}{4}$	-20397387126	2161683409874

alternate in sign (Figure 2), so E has a root in each of the five intervals they bound. A quintic has no others, and each of these roots is simple. All five lie in $(0, 3)$, where $D < 0$; hence $Y > 0$ and $X > 0$ there. This proves the count.

For the determinant, the two-by-two minor of Df in the variables (x, y) for the equations (f_2, f_3) is

$$M(z) = \det \frac{\partial(f_2, f_3)}{\partial(x, y)} = \tfrac{8}{5}z - \tfrac{2}{5}z^2 + \tfrac{1}{20000},$$

which is positive on $(0, 4)$. Differentiating along the curve, as in Section 5, gives at every root of E

$$\det Df = -\frac{M(z)\, E'(z)}{1250\, D(z)^2}.$$

At the five simple roots E' has the signs $+, -, +, -, +$ of the crossings, which gives the stated determinant signs. Stability of the first, third and fifth states is certified in exact rational arithmetic (`verify/certify_five.py`): each state has a rational isolating interval for its root of E, and on that interval the numerator and denominator of each Routh–Hurwitz quantity b_1, b_3, $b_1 b_2 - b_3$, written as a rational function of z along the curve, have no real root (Sturm counts), so its sign is its sign at the midpoint. States with $\det Df > 0$ have a positive real eigenvalue. □

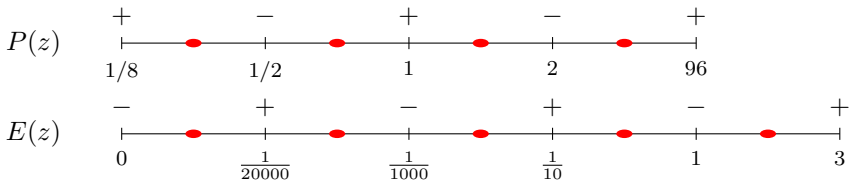


Figure 2: Sign diagrams, not to scale. Top: the quartic P of Section 5 changes sign four times on $(1/8, \infty)$, giving the four steady states of the seven-reaction network. Bottom: the quintic E of Section 6 changes sign five times on $(0, 3)$, giving the five steady states of the eight-reaction network. Dots mark the roots.

The coordinates are approximately $(2.68 \times 10^{-5},\ 29.80,\ 1.08 \times 10^{-5})$, $(9.35 \times 10^{-5},\ 8.554,\ 1.19 \times 10^{-4})$, $(0.00210,\ 0.3966,\ 0.01595)$, $(0.0814,\ 0.5690,\ 0.5591)$ and $(0.5964,\ 2.177,\ 2.176)$. These values play no part in the proof.

The quadratic coefficient matrix is

$$B = \begin{pmatrix} 2/5 & 0 & -1 \\ -2/5 & 1 & 0 \\ -2/5 & 0 & 0 \end{pmatrix}, \qquad \det B = -\tfrac{2}{5} < 0.$$

The system is sparse, so Theorem 2.1 does not apply to it directly. Its five nondegenerate states persist under every sufficiently small strict perturbation, which keeps $\det B < 0$, so those strict systems are regular five-state systems in the absorbing-box region, as Corollary 2.3 requires. The construction also explains where the example came from. The seven-reaction network obtained by deleting $X_2 \to X_1$ has four positive steady states and a negative quadratic determinant. Corollary 2.4 and the parity statement then force a fifth state in any small strict perturbation, and adding the single reaction $X_2 \to X_1$ at a small rate is enough to produce it.

7 Strict sharpness and exact reproducibility

In monomial order $(1, x, y, z, xy, xz, yz)$, a strict integer coefficient matrix attaining four positive steady states is

$$C = \begin{pmatrix} 1 & -1 & 1 & 101 & -801 & -101 & 101 \\ 101 & 1 & -1 & 1 & -801 & 101 & -1 \\ 1 & 301 & 1 & -101 & 101 & -1 & -1 \end{pmatrix}. \tag{7}$$

It is obtained by multiplying (4) by 100 and adding one unit in every prescribed coefficient direction. Its quadratic determinant is $776596 > 0$. Exact lexicographic elimination gives a squarefree quintic; all five roots are real, one negative and four positive. The latter lift to positive coordinates, with determinant signs $-, +, -, +$. Rational root isolation and interval sign checks certify the count and two stable states. This proves sharpness inside the strict class, rather than just on its boundary.

The accompanying verifier checks strict examples with $N = 0, 1, 2, 3, 4$, including the predicted determinant signs, degree sums, and exact inward or outward barrier inequalities. The three-state strict witness is a perturbation of [2, Example 3], not a separate new topology. Full coefficient matrices, exact root certificates and verification commands are at [1].

8 Formal verification

Both extreme examples are verified in Lean 4 with Mathlib [7, 8]. For each network the formal statement records the mass-action system as transcribed, reaction by reaction, from the reaction list (which appears in the Lean source as a comment; the reactions themselves are not formalised), proves that the stated Jacobian entries are the partial derivatives, and proves that the set of positive steady states consists of exactly four, respectively five, explicit points, each in a stated rational interval, with determinant signs $-, +, -, +$ and $-, +, -, +, -$. The proofs follow Sections 5 and 6: the rational curve, the sign changes of the eliminant, the bound on the number of roots of a polynomial, and the determinant identity. They use only the standard axioms of Lean's logic. Each formalisation was tested against planted errors (a wrong count, a wrong sign, a changed rate, a wrong Jacobian entry, a shifted root interval); all of the 10 and 11 planted variants respectively fail to compile, each with a Lean error in the mutated file. The stability counts, which rely on the Routh–Hurwitz criterion, are certified in exact rational arithmetic (`verify/verify_dichotomy.py` for Section 5, `verify/certify_five.py` for Section 6) but are not formalised. The general dichotomy is not formalised; its degree-theoretic steps use the standard properties of the Brouwer degree.

9 What remains open

The least number of reactions admitting five positive steady states is seven or eight. Eight is attained by Theorem 6.1. Seven is not excluded: the mixed volume of the steady-state supports

already reaches five for some seven-reaction networks. The construction of Section 6 starts from a seven-reaction network with four states and so cannot produce a seven-reaction example; we have not found one by other means.

The hypersurface $\Delta = 0$ and systems with vanishing coefficient slots require separate analysis for global dynamics and degree; the sparse upper bound and the finite-time-escape criterion are the stated exceptions. The four-state network obtained by deleting $X_2 \to X_1$ from Section 6 has $\det B < 0$ and an even count, so the parity statement genuinely needs the strict class. Extending the matrix B and the dichotomy beyond three species is a natural next step that this paper does not attempt.

Two other literatures meet these arguments. The barriers of Section 3 check bilinear functions at the corners of box faces. This is the vertex principle for multi-affine vector fields on rectangles, used for control by Belta and Habets [9] and extended to polynomial fields by Ben Sassi and Girard [10]; here it is applied on boxes that grow without bound, where only the quadratic part decides. Boundedness of mass-action systems can also follow from network structure alone: strongly endotactic networks are permanent [11]. That criterion does not involve the rate constants. The determinant Δ does. In the strict realisation each coefficient slot comes from its own reaction, so scaling the rates of the reactions on the diagonal of B moves Δ through zero in a single network, and for such a network no rate-independent criterion can decide boundedness. Classical dissipativity criteria for Lotka–Volterra systems [12] concern quadratic parts of the form $x_i(Ax)_i$, which differ from the reciprocal form (3).

Odd parity for dissipative networks without boundary equilibria is already part of the general degree framework of Conradi et al. [5]; odd parity alone is not a new claim here. The proof uses standard Perron–Frobenius and Brouwer degree principles, together with Bernstein's root bound; its proposed contribution is their explicit coefficient-level combination for this sign class, the sharp bound in one region, its connection with finite-time escape, and the two extreme examples.

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The question of why three steady states is the limit is Tang and Zhang's [2]. Two AI systems were used in this work under the author's direction. OpenAI Codex found the integer seven-reaction network of Section 5, proposed the determinant dichotomy and its proof, wrote the exact verifier and its certificates, and drafted the first version of the text. Claude Opus 5.5 (Anthropic) audited that draft and its proofs, found the eight-reaction five-state network and the elementary proofs of both extreme examples, wrote the Lean formalisation and the independent recheck, and revised the text. The author is responsible for the content of this preprint, which has not yet been independently reviewed.

Changes in version 2. After a Codex referee report: Corollary 2.3 now assumes $\Delta \neq 0$, which its proof through Theorem 2.1 needs; the stability of the five-state example, claimed in version 1 as exact but checked there only in 60-digit arithmetic, is now certified exactly (`verify/certify_five.py`); the description of the Lean statements says that the mass-action systems are transcribed, not derived from a formal reaction list; the Lean mutation harness accepts only errors located in the mutated file. Theorem 2.1 and the examples are unchanged.

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